



MINING DEALER BEST PRACTICE

Manufacture Oversize Hoist Bolt Kit

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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June 2016
1512-3.04-1406

1.0 Introduction

This improvement has been entered into the production line serviceable components Finning 495HR CRC Shovels and Drilling. Finning had to manufacture oversize bolts which have the function of extending the use of the Drum and Crown components.

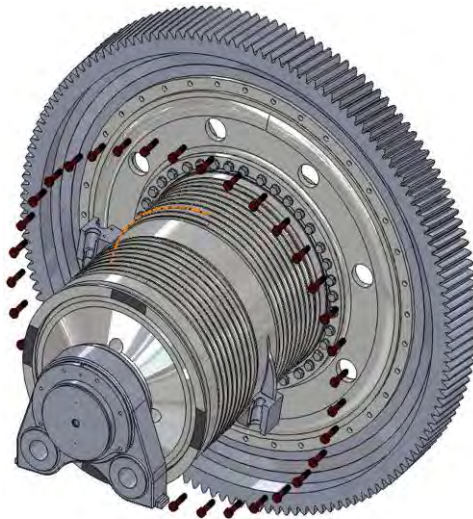
2.0 Best Practice Description

Generally in the assembling processes of the hoist drum (Drum P/N S006620 and Crown P/N E020030-01), the following pin is used: bolt P/N B006876-01 - Standard measurement (1.750") or bolt P/N B006876-02 – Standard measurement (1.810").

When the bolt B006876-01/02 cannot be used because of the bolt bore is out of line and will not fit, we need to manufacture an oversize bolt to reuse components.

Cross Reference

	Bucyrus	Caterpillar
Hoist Drum	S006620	1401502
Crown	E020030-01	1389171
Bolt	B006876-01	1271430
Bolt	B006876-02	



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Manufacture Over Size Hoist Bolt Kit

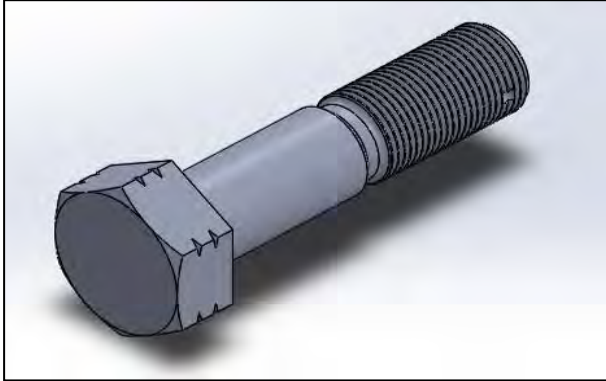
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3.0 Implementation Steps

A. STANDARD BOLT MANUFACTURING CAT PARTS PN B006876-01

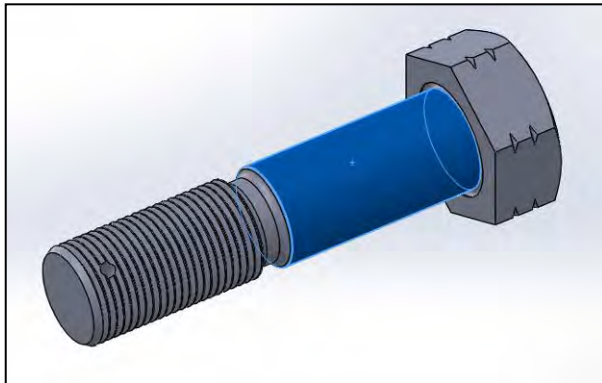


Picture 1

Bolt PN B006876	
Description	Measure
Head	2.750"
Body	1.750"
Thread	1.750 – 5 UNC
Bolt length	7.250"

Figure 1. This picture represents crown fixing bolt and drum, nominal size 1.750" +/- 0.002"

A. OVERSIZE BOLT FOR COMPONENTS SALVAGE



Picture 2

Bolt PN B006876- SM	
Description	Measure
Head	2.750"
Body	1.830"
Thread	1.750 – 5 UNC
Bolt length	7.250"

Figure 2, The oversize bolt with 1.830" nominal size for Drum and Crown assembly. The bolt is +0.080 larger than original diameter to assure it fits with the bolt bore.

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Manufacture Over Size Hoist Bolt Kit

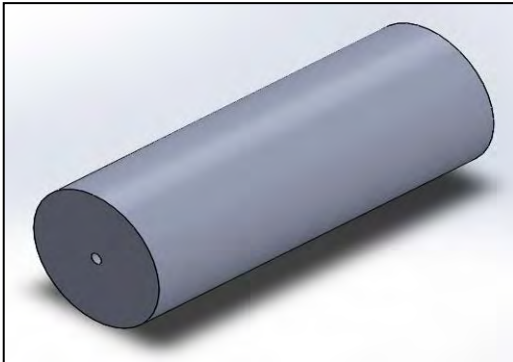
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A. STEPS OF THE MANUFACTURING PROCESS

1. Facing and turning point



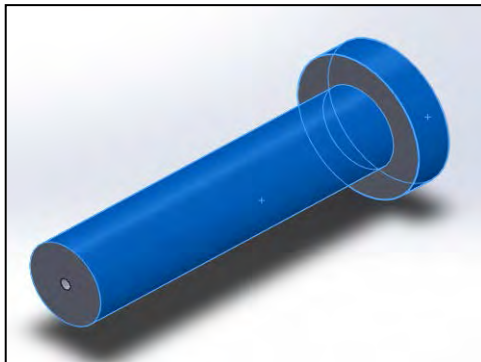
Material: Steel Bar AISI / SAE 4340

Hardness: 270-305 HB

Effort Creep > 70 (kg / mm²) **Traction Effort** 95 ^ -105 (Kg / mm²)

Dimensions: Diameter 3" x Length 8.250"

2. Rough and Machined Head and Body



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Descriptions:

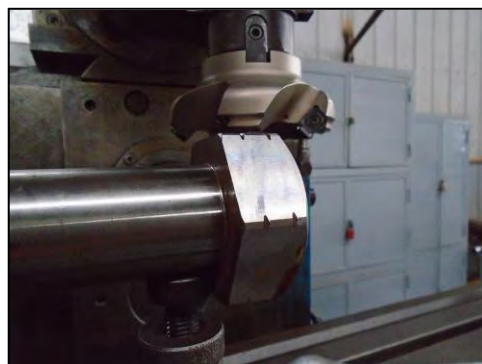
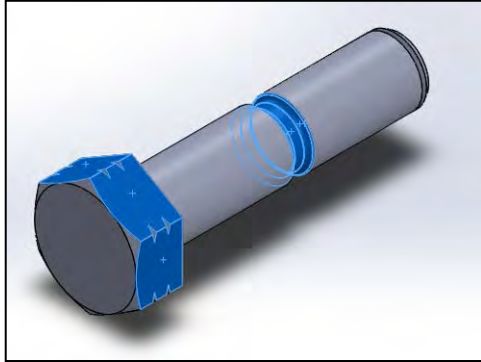
Equipment: Semi CNC lathe SN 71

RPM Used: 650-810 rpm

Feedrate: 0.27 mm / min

Inserts Used: TNMG 220408-MN (KENAMETAL) VNMG 160408-PM (SANDVIK)

3. Grinding Machining and Head Milling



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Descriptions:

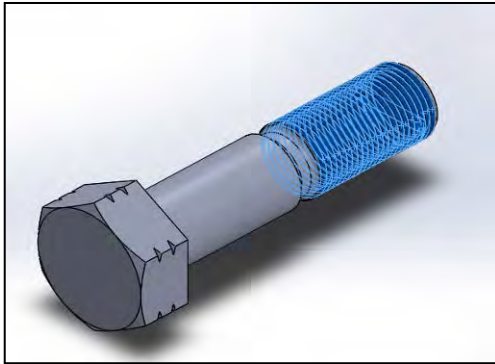
Equipment: Conventional milling and Semi CNC lathe SN 71

Used RPM: 400-450 rpm

Machining feed 0.6 mm / min

Inserts Used: SEXT 14M4AGSN-MN (KORLOY)

4. External thread



Description:

Equipment: Conventional Lathe SN 50

Used RPM: 125 rpm

Thread type: 1.75 - 5 UNC 2A

Feedrate: According Standardized Chart

Inserts Used: ERM22-N60 (KORLOY) QUALITY PC3030T 266RG-22 (SANDVIK)
QUALITY 1020

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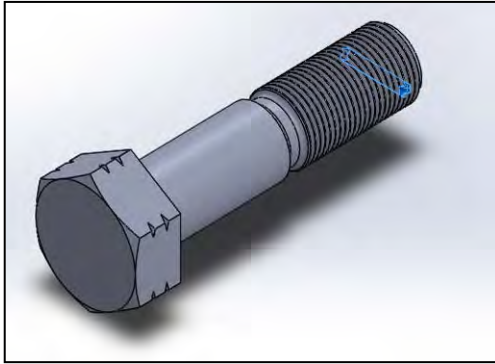
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5. Drilling of security hole



Descriptions:

Equipment: Radial Drill
 Used RPM: 250 RPM
 Type of wire: 0.12 mm / min.
 Used inserts: Bit 0.27 15% Cobalt
 Length: Trainee Drilling

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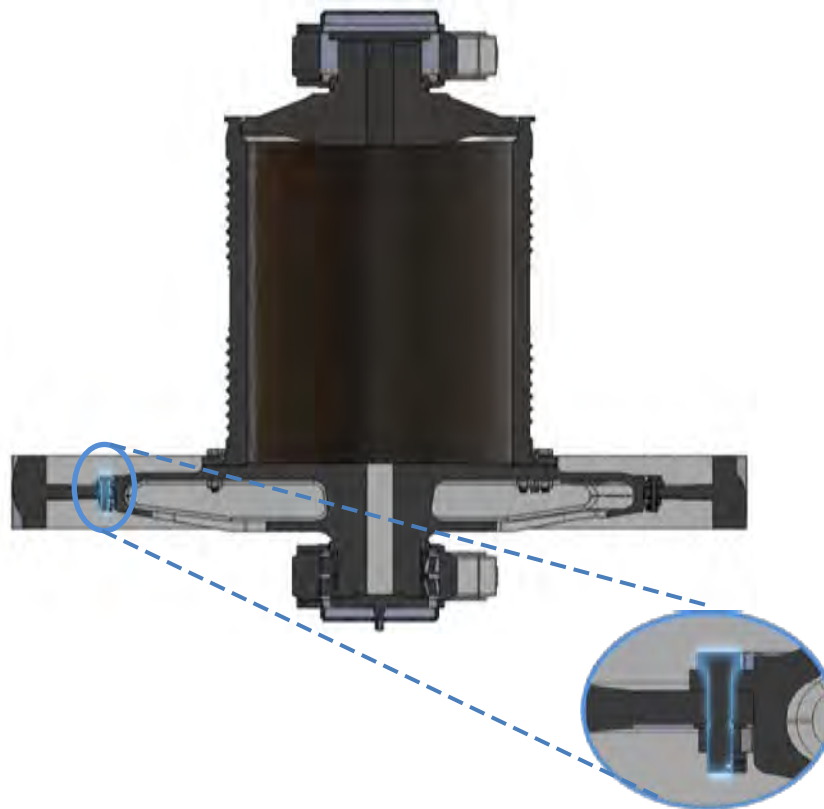
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6. Drum and crown ream and bolt installation

The bolt bore must be mechanized for the final measurements to be uniform (diameter bolt bore of drum = diameter bolt bore of spider).



Tools:	Manual Expandable Reamer
Reamer Range:	1.732" to 1.811"
Adjusting nominal bore diameter:	1.750" to 1.771"
Length:	Intern for both components



Hoist bolt installed

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4.0 Benefits

Reuse drum/spider assembly when:

- The bolt (part number B006876-01/02) cannot be used.
- The measurement bolt bore is out the maximum permissible range.

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5.0 Resources Required

Machinery and equipment Tools

- Conventional Lathe SN 50
- Semi CNC lathe SN 71
- Conventional milling
- Radial Drill

Materials

- AISI steel bar / SAE4340
- Inserts Machining
- Drill Center
- Cobalt Drill

6.0 Supporting Attachments / References

- Reference Drawing PN B006876-01

EDITOR'S NOTE: B006876-01 has been converted to an AA part type. Check NextGen SIS for the latest part information.

7.0 Related Best Practices

- N/A

8.0 Acknowledgements

- Finning – CRC Shovels and Drill.

Recognition

Team Rafael Nilo/ Machine Shop Supervisor
 Johny Perez/Technical Supervisor
 Nelson Troncoso/Welding Supervisor
 Daniel Gajardo/Mechanical Supervisor
 Rodrigo Vicencio/ Operation Chief

Sponsor Sebastián Gómez/ CRC Manager – Expanded Product Line

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